

Data Preparation & Power Query Transformations

Airline Flight Operations & Delay Analysis

Tool: Microsoft Power BI / Power Query **Dataset:** Airline Flight Delays **Period:** 2015

1. Purpose

This document summarizes the data preparation process followed for the Airline Flight Operations & Delay Analysis project. The objective was to convert the raw CSV files into a consistent, analysis-ready structure before creating the data model, DAX measures, and Power BI dashboard.

2. Source Data

Table	Purpose
flights	Central flight-level operational data used for delay, cancellation, duration, and flight-volume analysis.
airlines	Reference table containing airline names and IATA codes.
airports	Reference table containing airport codes, names, cities, states, and countries.

3. Power Query Preparation Process

Data Loading: The three CSV datasets were imported into Power BI through Power Query and kept as separate tables to preserve the source structure.

Data Profiling: The tables were reviewed to understand columns, data types, missing values, identifiers, delay fields, cancellation fields, and timing-related fields.

Data Type Validation: Fields were reviewed so that text, numeric, date/time, and identifier columns could be used correctly for filtering, aggregation, relationships, and calculations.

Missing Value Review: Null values were reviewed according to business meaning. A blank value was not automatically treated as an error; for example, cancellation-related fields may naturally be blank for flights that were not cancelled.

Identifier Preparation: Airline and airport IATA codes were reviewed because they are used to connect flight records with the Airlines and Airports reference tables.

4. Table Preparation

The **Flights** table was prepared as the central operational table containing the records used for flight volume, delay, cancellation, duration, and delay-contributor analysis.

The **Airlines** table provides descriptive airline information used for airline filtering and performance analysis.

The **Airports** table provides descriptive airport information used for airport-level analysis using meaningful names, cities, and states.

5. Preparation for the Power BI Model

After preparation, the tables were structured for a fact-and-dimension style model. Flights acts as the central fact table, while Airlines and Airports provide descriptive reference information.

Table	Role
Flights	Central fact table containing flight-level operational records.
Airlines	Reference/dimension table for airline information.
Airports	Reference/dimension table for airport information.

6. Data Quality Validation

Before dashboard development, the prepared data was reviewed to confirm that the required tables and columns were available, data types were appropriate, identifiers could support relationships, and delay and cancellation fields were suitable for analysis. KPI results and dashboard filtering were checked during validation.

7. Power Query and DAX — Different Responsibilities

Power Query was used for preparing the source data and making it suitable for modeling. **DAX** was then used to create analytical measures and KPIs such as Total Flights, Delayed Flights, Delay Rate, Cancelled Flights, and Average Flight Duration.

8. Final Outcome

The preparation process produced an analysis-ready structure that supported the final Power BI Executive Overview dashboard, including monthly analysis, airline analysis, origin-airport analysis, delay severity, delay contributors, and cancellation analysis.

9. Conclusion

Power Query provided the foundation for the project by loading, reviewing, validating, and preparing the source datasets before modeling and dashboard development.

CSV Source Data → Power Query → Data Model → DAX Measures → Power BI Dashboard → Insights

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